

GE Healthcare

Prospective Respiratory Gating Option Installation Service Documentation



5321869-1EN

Rev 1

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LANGUAGE

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WARNING (EN)	• This Service Manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.
HOIATUS (ET)	• Käesolev teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
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ATENÇÃO (PT-PT)	• Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutra linguagem, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
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ATENCION (ES)	• Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
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To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

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All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
1	11/13/08	Initial release.

Table of Contents

Chapter 1

Prospective Respiratory Gating Installation..... 17

Section 1.0

Introduction 17

1.1 Theory..... 17

1.2 Overview..... 17

Section 2.0

Material List 17

Section 3.0

Hardware Installation 18

3.1 Prerequisite..... 18

3.2 Cable Connection 18

Section 4.0

Software Installation and Functional Test 19

4.1 Prerequisite..... 19

4.2 Installation and Test..... 19

Section 5.0

Finish Installation..... 20

5.1 Install Rating Plate..... 20

5.2 Process Product Locator Card..... 20

Chapter 1

Prospective Respiratory Gating Installation

Section 1.0 Introduction

1.1 Theory

The Prospective Respiratory Gating (PRG) option is to provide the CT system with respiratory gating scan. The prospective respiratory trigger is designed to aid in assessment of tumor location by providing a consistent scan start synchronized with the respiratory cycle for radiation treatment planning.

The patient's respiratory cycle is recorded as a respiratory motion waveform via an external respiratory monitoring device (for example RPM, Real-Time Position Management). CT data acquisition is automatically triggered at a desired threshold in the patient's respiratory cycle. Two types of scan mode Axial and Helical can be triggered by the respiratory trigger, called Axial-R and Helical-R.

1.2 Overview

This option kit only contains RPM to Gantry Cable (5199717), but this document gives an overall description of the PRG option installation, including prerequisites.

Section 2.0 Material List

Verify that the contents of this option kit are correct according to the Material List below.

ITEM	PART NO.	DESCRIPTION	QTY.
1	5199717	RPM to Gantry Cable	1
2	5321869-1EN	Prospective Gating Option installation Instruction	1

Section 3.0 Hardware Installation

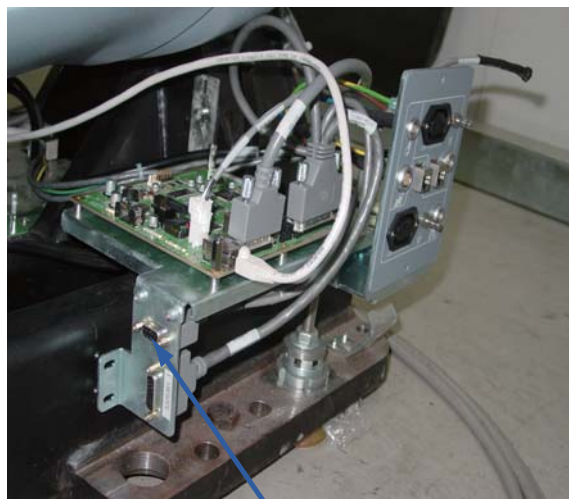
3.1 Prerequisite

The following hardware must be installed prior to this option kit installation. Refer to the relevant option installation manual.

- 1.) Gantry Accessory I/F Hardware with IPC (Integrated Peripheral Control) board. This is sold separately, and is NOT part of this option kit.
- 2.) Accessory AC Outlet in console must be installed. This is NOT part of this option kit.
- 3.) The RPM (Remote Patient Monitoring) system (Camera) must be installed and functional. This RPM system is sold separately, and is NOT part of this option kit.

3.2 Cable Connection

Connect the RPM to Gantry Cable (5199717) from IPC bracket to RPM IF box. Please refer to Figure 1-1.



From upper
connect on
IPC bracket



To RPM
IF Box

Figure 1-1 RPM to Gantry Cable (5199717) Connection

Section 4.0

Software Installation and Functional Test

4.1 Prerequisite

Software version 08BW44.1 or later is required for installation of PRG Option Key.

4.2 Installation and Test

- 1.) Install the respiratory gating option software using the option key B70222RT.
- 2.) Work with CT Scanner Technologist or CT Applications Specialist to perform respiratory gating scans. Refer to the operator manual (Learning and Reference Guide), Respiratory Triggering chapter for details.

Section 5.0

Finish Installation

5.1 Install Rating Plate

Attached the Rating plate on the rear of the OC as shown in illustration below. Please refer to [Figure 1-2](#)

OC Rear Side

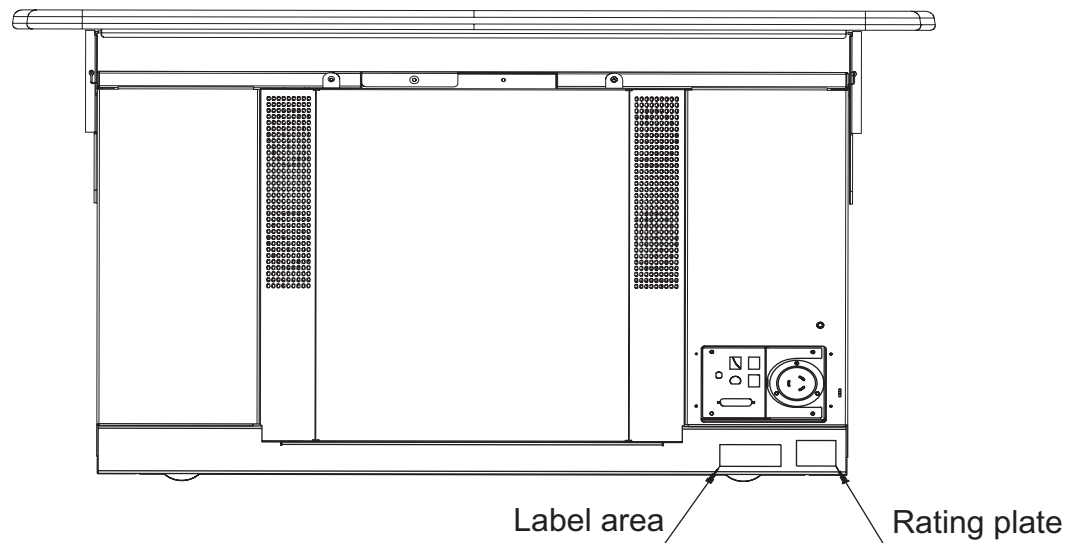


Figure 1-2 Rating Plate Location

5.2 Process Product Locator Card

Process the product locator card according to the instruction in your country.

www.gehealthcare.com



Imagination at work